

Quadratic factors — 3.4

Example: $\int \frac{3x-1}{x^3+x} dx$ $x^3+x = x(x^2+1)$

$$\frac{3x-1}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$3x-1 = A(x^2+1) + (Bx+C)x = Ax^2 + Bx^2 + Cx + A$$

$$\begin{cases} A+B=0 \\ C=3 \\ A=-1 \end{cases} \Rightarrow B=1$$

$$\int \frac{3x-1}{x^3+x} dx = - \int \frac{1}{x} dx + \int \frac{x+3}{x^2+1} dx$$

$$\begin{aligned} u &= x^2+1 \\ du &= 2x dx \end{aligned}$$

$$- \ln|x| + \int \frac{x}{x^2+1} dx + \int \frac{3}{x^2+1} dx = - \ln|x| + \frac{1}{2} \int \frac{1}{u} du + 3 \int \frac{1}{1+x^2} dx$$

$$= - \ln|x| + \frac{1}{2} \ln|x^2+1| + 3 \arctan(x) + C$$

Example: Find $\int \frac{2x^2 + 10x + 13}{x(x^2 + 6x + 13)} dx$

$$\frac{2x^2 + 10x + 13}{x(x^2 + 6x + 13)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 6x + 13}$$

$$2x^2 + 10x + 13 = A(x^2 + 6x + 13) + (Bx + C)x$$

$$x=0: 13 = 13A \Rightarrow A=1$$

$$x=1: 25 = 20A + B + C = 20 + B + C \Leftrightarrow B + C = 5$$

$$x=-1: 5 = 8A + B - C = 8 + B - C \Leftrightarrow \underline{B - C = -3}$$

$$2B = 2 \Rightarrow B = 1$$

$$\Rightarrow C = 4$$

$$\int \frac{2x^2 + 10x + 13}{x(x^2 + 6x + 13)} dx = \int \frac{1}{x} dx + \int \frac{x+4}{x^2+6x+13} dx$$

Completing the square: $x^2 + 6x + 13 = x^2 + 6x + 9 + 4$
 $= (x+3)^2 + 4$

$\rightarrow = \ln|x| + \int \frac{x+4}{(x+3)^2 + 4} dx$ $u = x+3 \Leftrightarrow x = u-3$
 $du = dx$

$$= \ln|x| + \int \frac{u+1}{u^2+4} du = \ln|x| + \int \frac{u}{u^2+4} du + \int \frac{1}{u^2+4} du$$

$$= \ln|x| + \frac{1}{2} \ln|u^2+4| + \frac{1}{2} \arctan\left(\frac{u}{2}\right) + C$$

$$= \ln|x| + \frac{1}{2} \ln|(x+3)^2+4| + \frac{1}{2} \arctan\left(\frac{x+3}{2}\right) + C$$

$$= \ln|x| + \frac{1}{2} \ln|x^2+6x+13| + \frac{1}{2} \arctan\left(\frac{x+3}{2}\right) + C$$

Example: We decompose the rational function

$$\frac{10x^2 + 8}{(x-2)^2(x^2+6x+13)(x^2+1)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{Cx+D}{x^2+6x+13} + \frac{Ex+F}{x^2+1} + \frac{Gx+H}{(x^2+1)^2}$$

Exercise: Try integrating $\int \frac{3x+10}{(x^2+6x+13)^2} dx$.